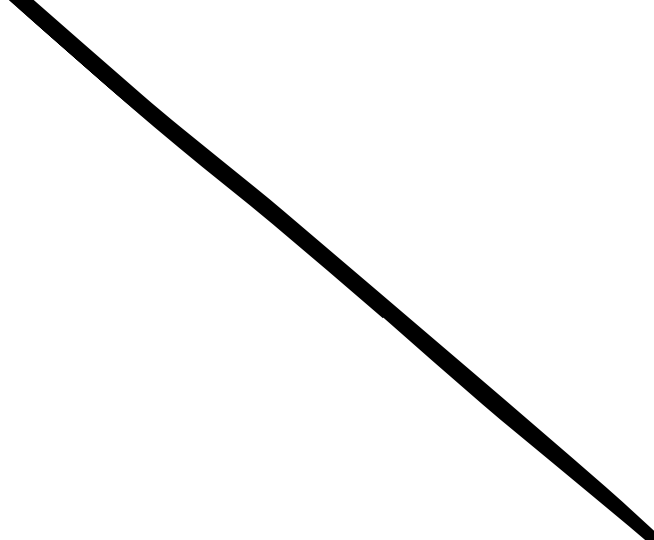




IEEE 802 LMSC and Ethernet

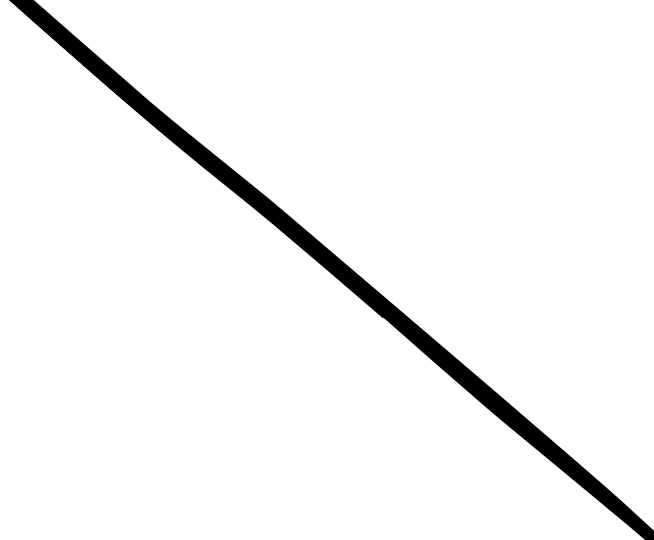
The IEEE 802 LAN/MAN Standards Committee

Group 21
陳宥儒 陳柏勳





IEEE 802 LMSC



Established in Feb 1980.

Committee name based on IEEE series committee numbers

- Not related to establishment date
- Aiding memory

Sets standards for Local Area Networks (LAN), Metropolitan Area Networks (MAN) and High Speed Networks

- <http://standards.ieee.org>
- <http://www.ieee802.org>

Organized into numerous Working Groups responsible for task advancement



IEEE 802 Working Groups (1/3)

Working Group	Standardizing	Status
IEEE 802.1	LANs and High-Level Interfaces (HLI) above OSI Layer 2	Active
IEEE 802.2	LLC (Logical link control)	Disbanded
IEEE 802.3	Ethernet	Active
IEEE 802.4	Token bus	Disbanded
IEEE 802.5	Token Ring MAC layer	Disbanded
IEEE 802.6	MANs (DQDB)(Distributed-queue dual-bus)	Disbanded
IEEE 802.7	Broadband LAN using Coaxial Cable	Disbanded
IEEE 802.8	Fiber Optic Technical Advisory	Disbanded

IEEE 802 Working Groups (2/3)

Working Group	Standardizing	Status
IEEE 802.9	Integrated Services LAN (ISLAN or isoEthernet)	Disbanded
IEEE 802.10	Interoperable LAN Security	Disbanded
IEEE 802.11	WLAN (Wireless Local Area Networks)	Active
IEEE 802.12	100BaseVG	Disbanded
IEEE 802.13	Reserved for Fast Ethernet development	Unused
IEEE 802.14	Cable modems	Disbanded
IEEE 802.15	WPAN (Wireless Personal Area Network)	Active
IEEE 802.16	Broadband Wireless Access	Hibernating

IEEE 802 Working Groups (2/3)

Working Group	Standardizing	Status
IEEE 802.17	Resilient packet ring	Hibernating
IEEE 802.18	Radio Regulatory advisory	Active
IEEE 802.19	Wireless Coexistence advisory	Active
IEEE 802.20	Mobile Broadband Wireless Access	Disbanded
IEEE 802.21	Media Independent Handoff	Disbanded
IEEE 802.22	Wireless Regional Area Network	Disbanded
IEEE 802.23	Emergency Services Working Group	Disbanded
IEEE 802.24	Smart Grid and vertical applications	Active

IEEE 802 and OSI Model

802.1:

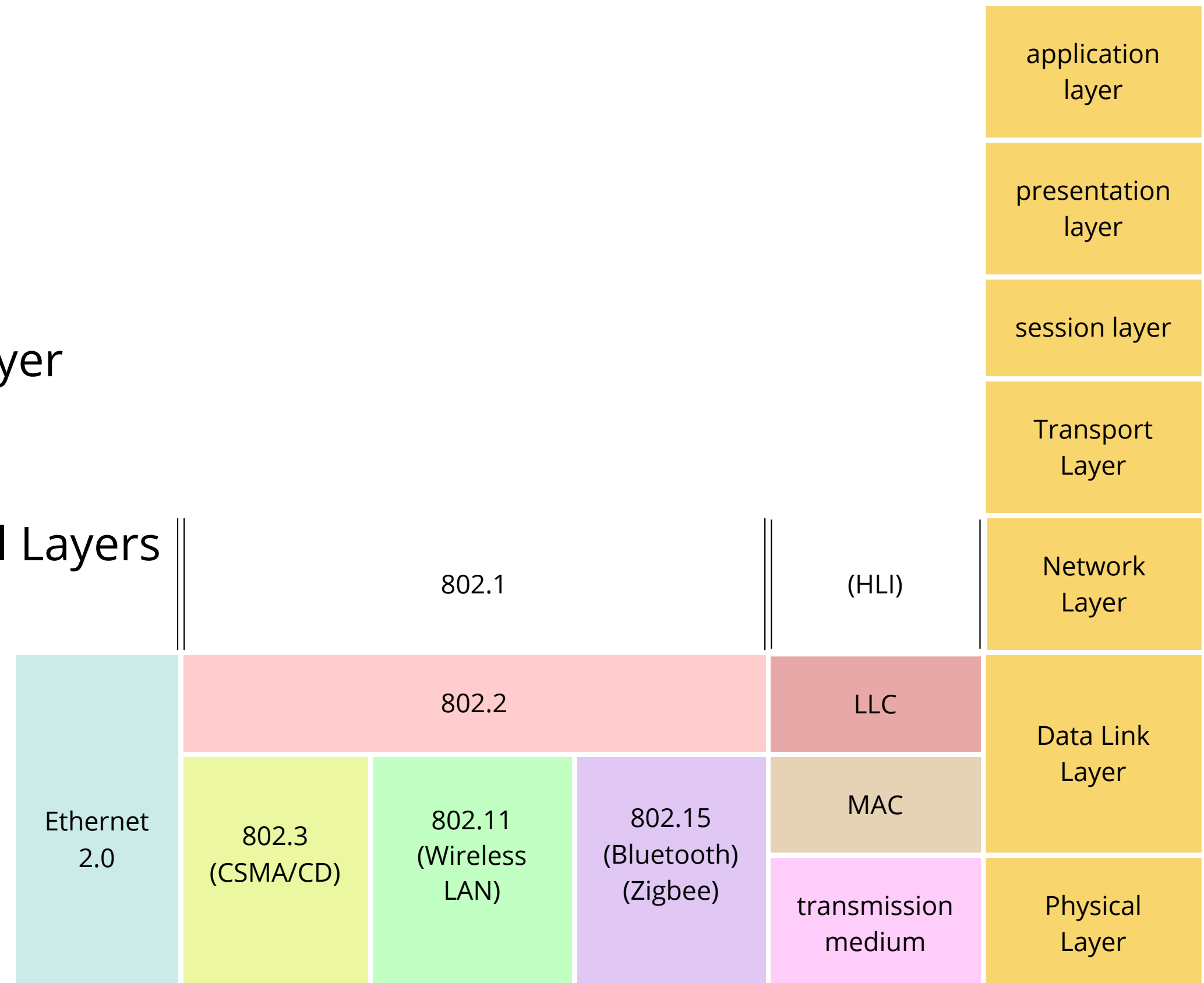
- High-Level Interface (HLI)

802.2:

- Corresponds to the LLC sublayer

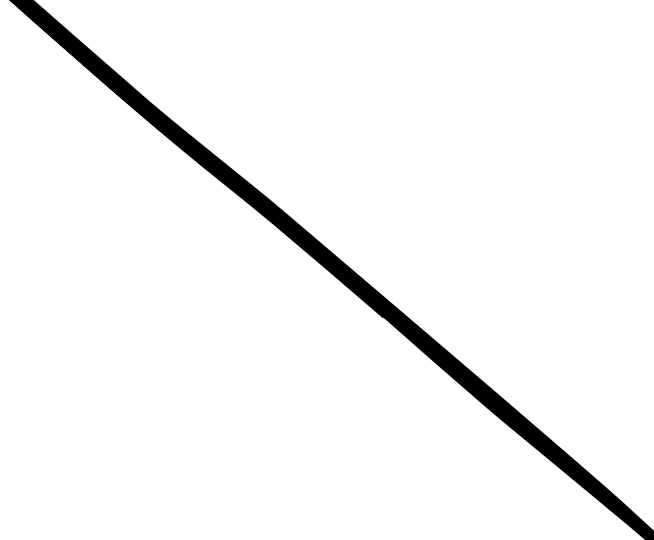
802.3 ~ 802.24:

- Correspond to MAC + Physical Layers





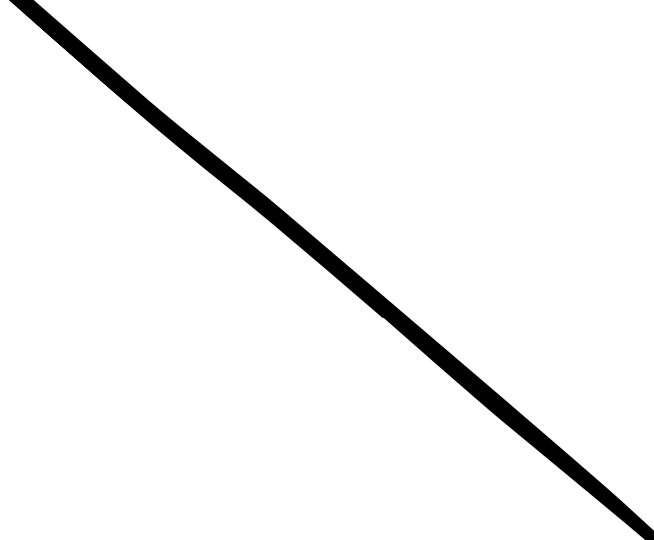
Typical LAN Architecture



- ◆ IEEE 802.3, 802.4, 802.5 are the three main wired LANs
 - ◆ 802.3 is most widely used
 - ◆ 802.3 uses CSMA/CD
 - ◆ 802.11: CSMA/CA
 - ◆ 802.15: Bluetooth
 - ◆ Zigbee is based on 802.15
- 



Ethernet Standard Introduction



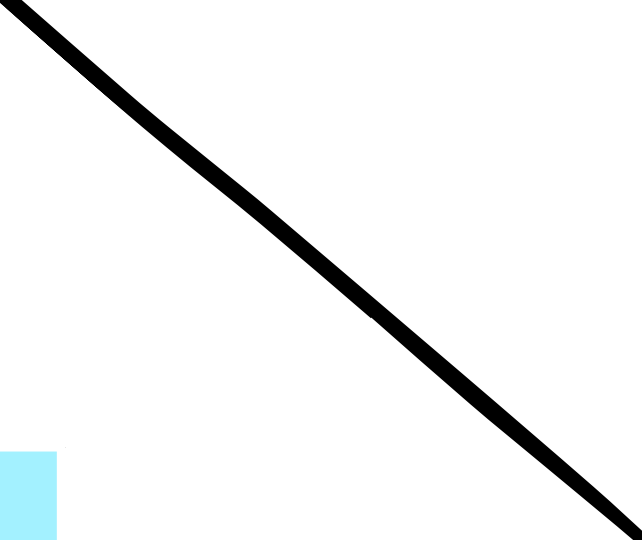
Currently the most commonly used LAN technology

Open standard: compatible with any manufacturer's equipment

- 1980: DIX 1.0
- 1982: DIX 2.0
- IEEE 802.3 released in 1985



IEEE 802.3

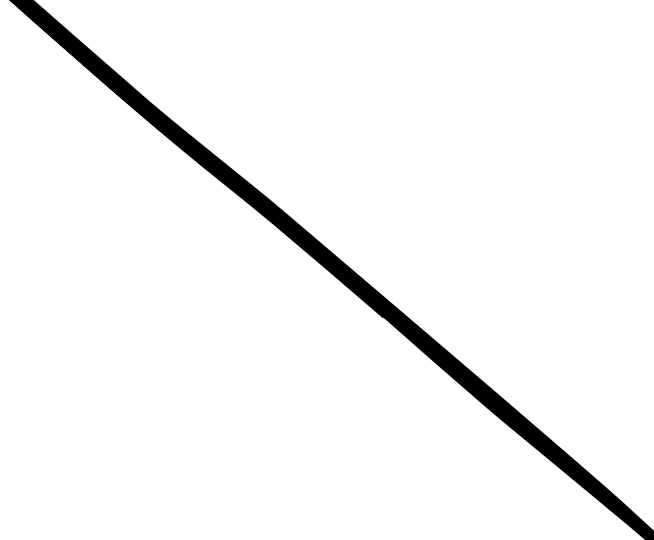


File name	Specifications
802.3a-1985	10Base2
802.31-1990	10BaseT
802.3j-1993	10BaseF
802.3u-1995	100BaseT 100BaseF
802.3y-1997	100BaseT2
802.3x-1997	100Mbps full-duplex
802.3ab-2001	1000BaseT

File name	Specifications
802.3z-1998	1000BaseX
802.3ae-2002	10Gbase(Full duplex, collision-free)
802.3af-2003	Power over Ethernet
802.3an-2006	10Gbase T
802.3ba-2010	40G/100G Ethernet
802.3bs-2017	400G Ethernet



Ethernet 2.0 vs IEEE 802.3



Both define Ethernet frames for data transmission

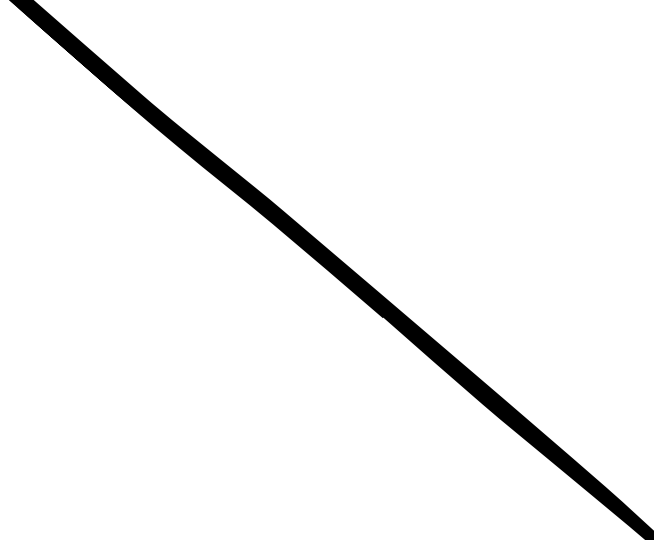
Ethernet 2.0 uses EtherType field

IEEE 802.3 uses Length field

Today's systems support both formats for compatibility

Both use CSMA/CD protocol





THANK YOU

